IT-IM 05

DATA WAREHOUSING AND DATA MINING

UNIT 1 – PART 2: DATA WAREHOUSE ARCHITECTURE

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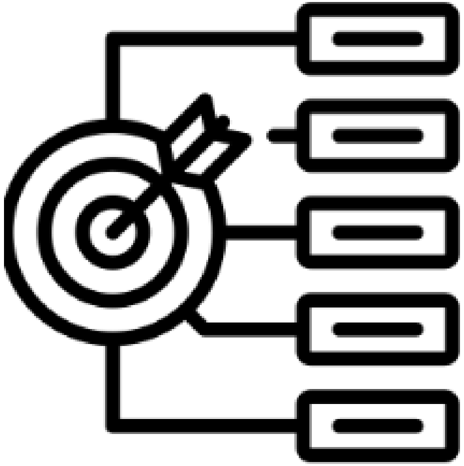
SUBJECT TEACHER

Learning Objectives

At the end of the lesson, the students will be able to:

- differentiate the two data warehouse architectures by correctly identifying their characteristics and appropriate applications;

- identify the different data warehouse schemas by correctly recognizing their structures and components.

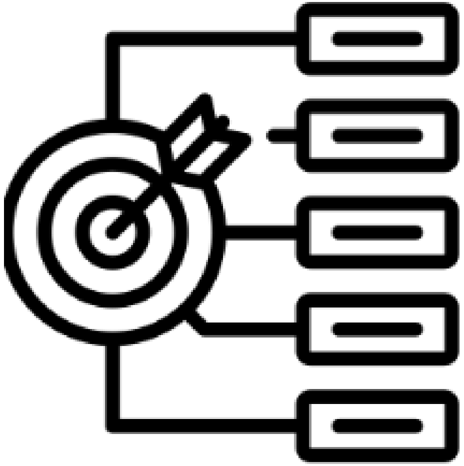


Learning Objectives

At the end of the lesson, the students will be able to:

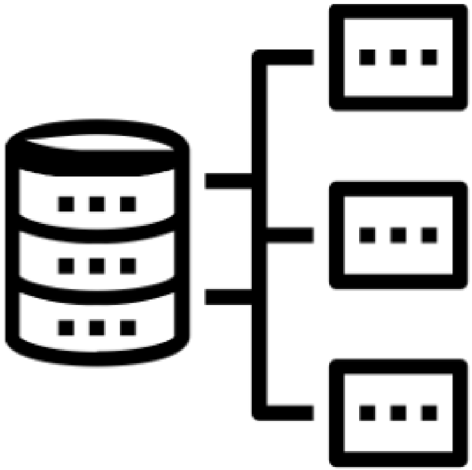
- explain the normalized and dimensional approaches by correctly describing the steps on each;

- perform normalization and dimensional modeling by correctly applying the concepts in designing relational tables as well as fact and dimension tables.



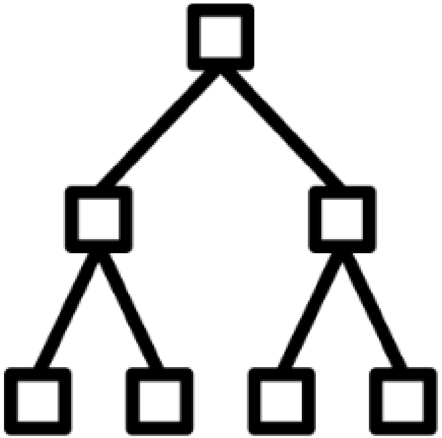
Data Warehouse Architecture

- Normalized Approach
- Dimensional Approach



Normalized Approach

- Normalized Data Warehouse
  - Based on the relational data model
  - Utilizes normalized table to reduce redundancy and ensure integrity
  - Ideal for:
    - When consistency and accuracy of data need to be preserved
    - Supporting complex data transformations
    - Complying on strict data quality standards

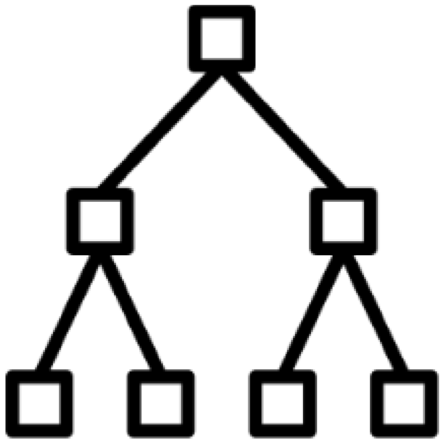


Normalized Approach

- Inmon Architecture
  - ☐ Developed by Bill Inmon
  - ☐ Focuses on a centralized and integrated approach in data warehousing

- Enterprise Data Warehouse (EDW)
  - ☐ a centralized repository that stores integrated and subject-oriented data

- Top-down approach
  - ☐ starts with EDW then into data marts



Inmon Model

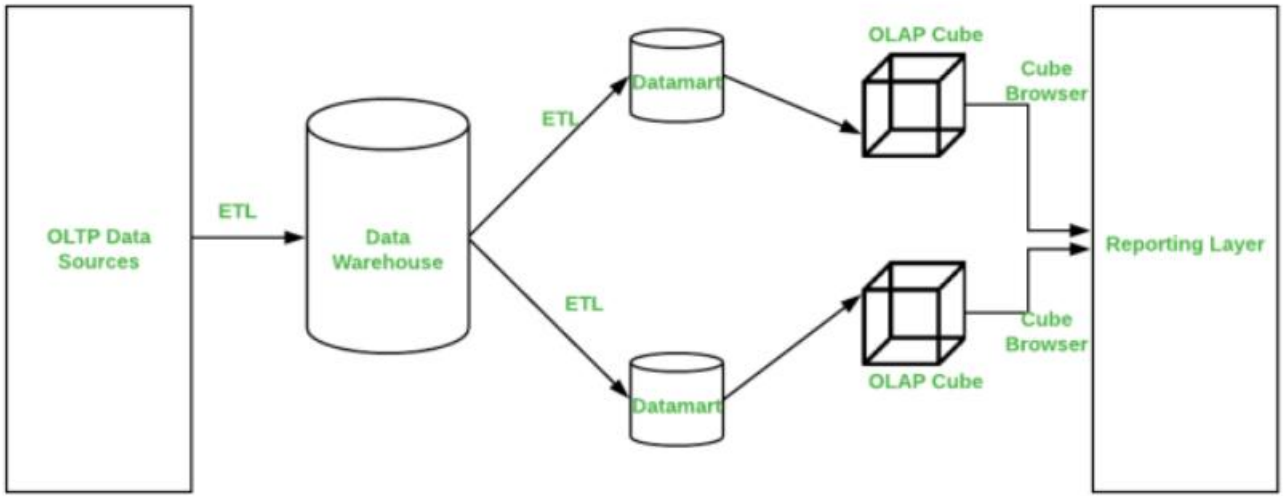


Image source: https://www.geeksforgeeks.org/difference-between-kimball-and-inmon/

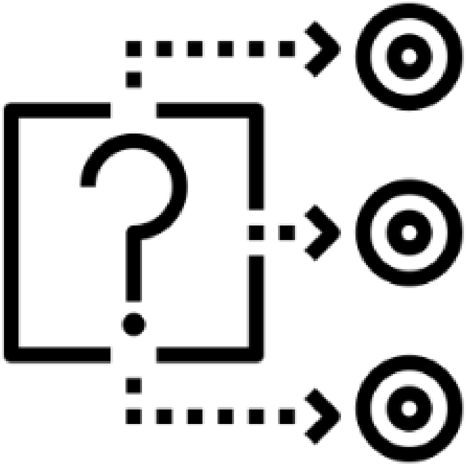
Pros and Cons

More proficient, dependable, and scalable

May require more complex queries, longer processing time, and higher technical skills

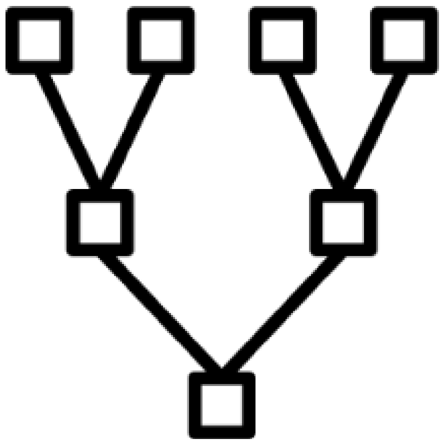
Reasons for Normalization

- Data Integrity
- Flexibility and Scalability
- Integration
- Reduced Data Anomalies



Dimensional Approach

- Dimensional Data Warehouse
  - ☐ Consists of facts tables and dimension tables
  - ☐ Fact tables: store quantitative measures
  - ☐ Dimension tables: store descriptive attributes
  - ☐ Ideal for:
    - ☐ fast and flexible reporting
    - ☐ ad-hoc analysis
    - ☐ data visualization

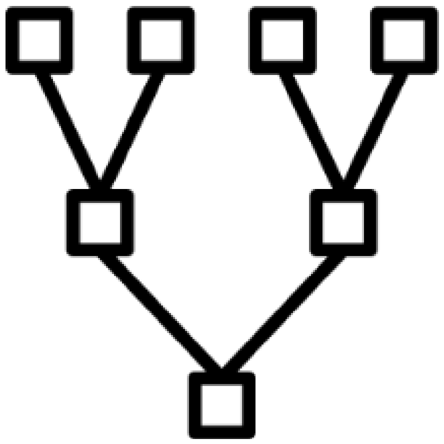


Dimensional Approach

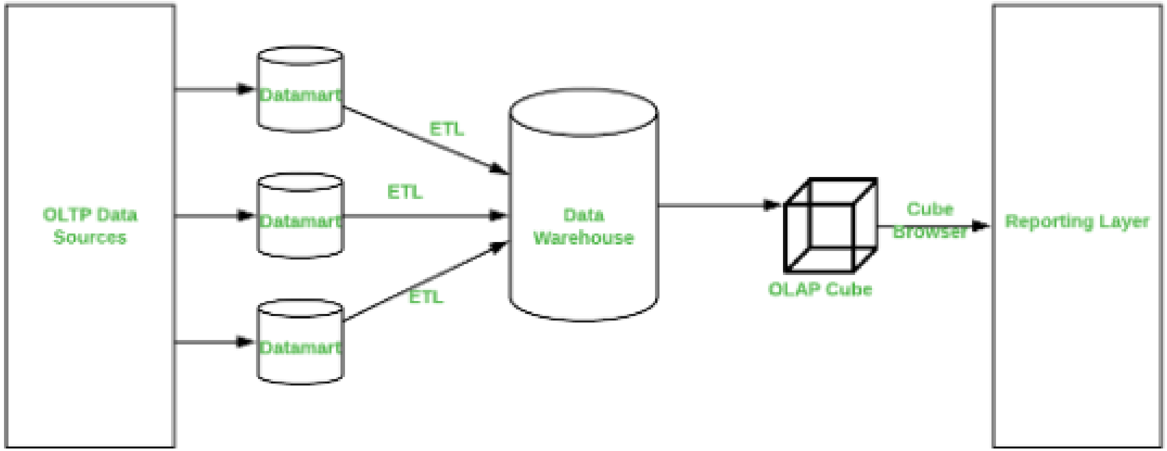
- Kimball Architecture
  - Developed by Ralph Kimball
  - Focuses on organizing data into facts and dimensions (**dimension model**)

- Bottom-up approach
  - Starts with individual data marts then integrating them into an enterprise data warehouse

- Star, Snowflake, Galaxy Schema



Kimball Model



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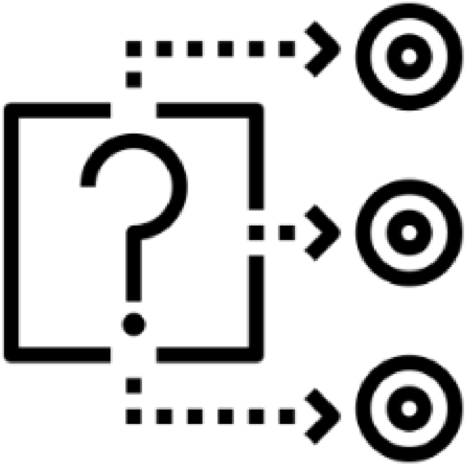
Pros and Cons

Easier to comprehend, utilize, and maintain

May require more data storage and there is a possibility for data duplication and decrease data quality

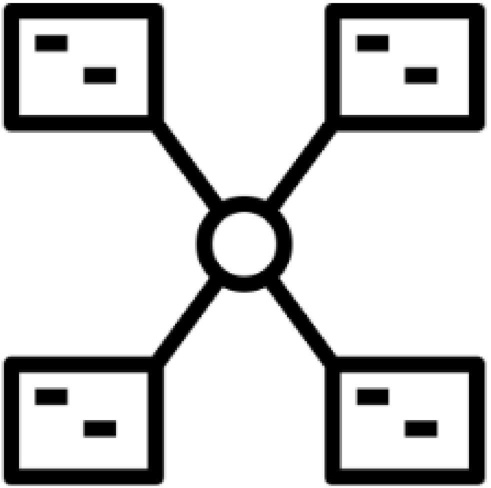
Reasons for Denormalization

- Query Performance
- User-Friendly
- Reduced Join Complexity
- Analytical Efficiency



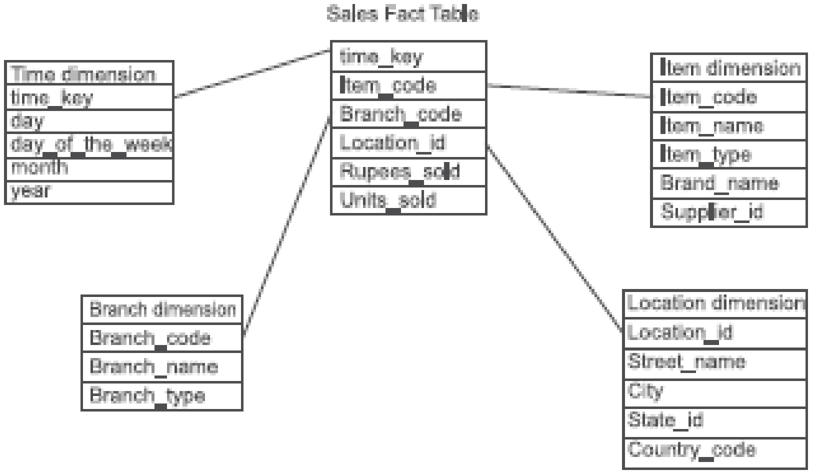
Schema

- Star Schema
- Snowflake Schema
- Galaxy Schema



Star Schema

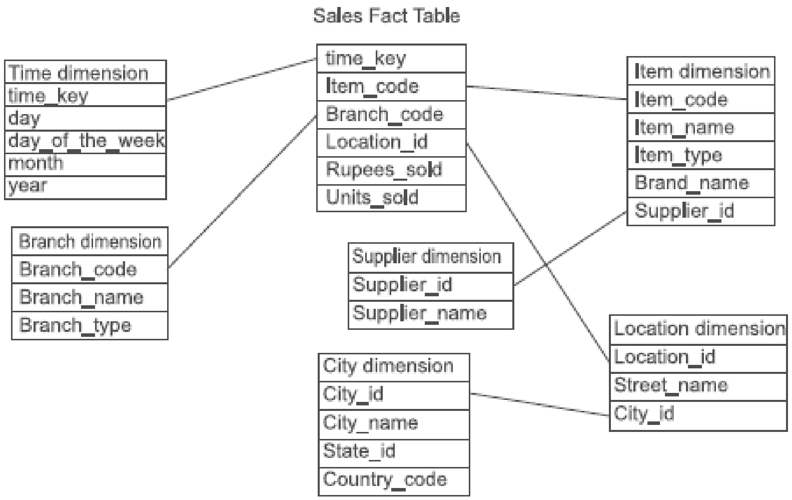
- Simplest schema
- Fact table is in the center
- Dimension tables are the nodes of the star
- Usually fact tables are normalized while dimensions are denormalized



Source: Bathia, Parteek (2019). Data Mining and Data Warehousing. Cambridge University Press

Snowflake Schema

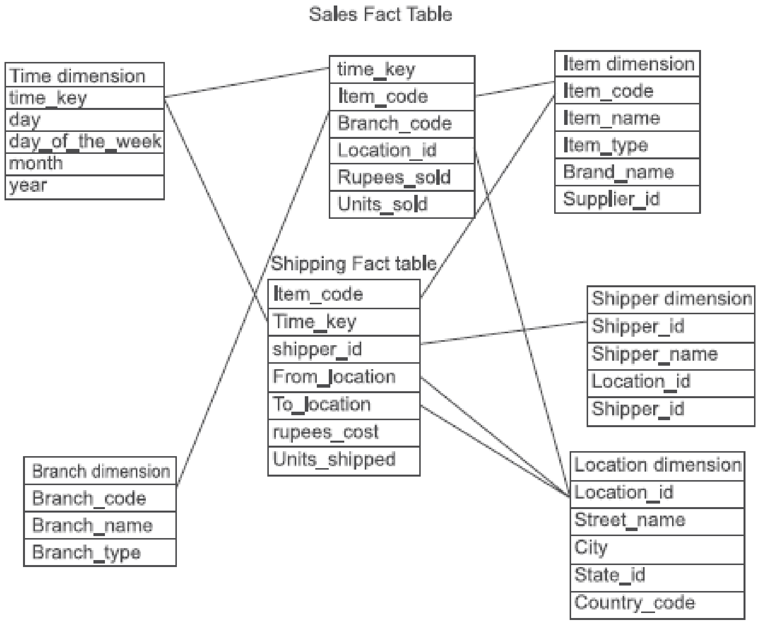
- The main difference between the star schema and the snowflake schema is that in snowflake, there are dimension tables that are normalized.



Source: Bathia, Parteek (2019). Data Mining and Data Warehousing. Cambridge University Press

Galaxy Schema

- The main difference between snowflake and galaxy is that galaxy consists of multiple fact tables



Source: Bathia, Parteek (2019). Data Mining and Data Warehousing. Cambridge University Press

Comparison

Aspect	Inmon (Normalized)	Kimball (Denormalized)
Schema Design	Centralized, normalized schema	Decentralized, dimensional schema (star/snowflake)
Data Redundancy	Low, minimal redundancy	High, intentional redundancy
Data Integrity	High, due to normalization	Moderate, managed through ETL processes
Query Performance	May be slower due to complex joins	Optimized for high performance
Ease of Use	Requires more technical understanding	User-friendly, intuitive for business users
Flexibility	High, easier to adapt to new requirements	Moderate, designed for specific business processes

Source: https://www.linkedin.com/pulse/understanding-kimball-inmon-data-warehouse-omar-khaled

Scenarios

Inmon Approach:

- **Centralized Data Warehouse**: A retail company integrates all sales, customer, and inventory data into a centralized, normalized warehouse.
- **Data Marts**: Specific data marts are created for departments like sales, marketing, and inventory, each drawing from the central warehouse.

Source: https://www.linkedin.com/pulse/understanding-kimball-inmon-data-warehouse-omar-khaled-c9ksf#

Kimball Approach:

- **Sales Data Mart**: The same retail company starts with a sales data mart using a star schema to provide quick insights into sales performance.
- **Customer and Inventory Data Marts**: Additional data marts are created for customer and inventory analysis, each designed for performance and ease of use.

References

- Bathia, Parteek (2019). Data Mining and Data Warehousing. Cambridge University Press
- Khaled, Omar (2024). Kimball and Inmon Data Warehouse Architectures. LinkedIn. https://www.linkedin.com/pulse/understanding-kimball-inmon-data-warehouse-omar-khaled-c9ksf#
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- Difference between Kimball and Inmon. GeeksForGeeks. https://geeksforgeeks.org/difference-between-kimball-and-inmon/